MILITARY
SPECIFICATION

MIL-T-12612(SigC) 8
2 April 1953
Superseding
U. S. Army Specificati
No. 74-115
14 April 1948

THEODOLITE MOUNT ML-180-( )

1. SCOPE

1.1 This specification covers one type of equipment designated as Theodolite Mount ML-180-( ) (see 6.4). This mount contains provisions for adjusting the height of the theodolite, protecting the theodolite from the weather when not in use and for supporting a lighted shelf for recording data.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND DRAWINGS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

[illegible]-B-591	Boxes; Fiberboard, Wood-Cleated (For Domestic Shipment)
NN-B-601	Boxes; Wood-Cleated-Plywood (For Domestic Shipment)
NN-B-621	Boxes; Wood, Nailed and Lock-Corner
LLL-B-631	Boxes; Fiber, Corrugated (For Domestic Shipment)

MILITARY

JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed
MIL-B-107	Boxes, Wood, Wirebound (Overseas Type)
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior
MIL-P-116	Preservation, Methods of
JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistant

Source: https://assist.dla.mil -- Downloaded: 20260-17T23:05Z
Check the source to verify that this is the current version before use.

U. S. ARMY

19-10	Filler, Interior Packing, Fiberboard, Corrugated Single and Double Faced, For General Packing Purposes

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking of Shipments

DRAWINGS

SIGNAL CORPS

SC-D-15914	Moisture-Resistance Test Cycle for Ground Signal Equipment
SC-DL-86479	Theodolite Mount ML-180-( )

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Preproduction samples.-

3.1.1 Quantities.- After award of contract, two preproduction samples of the complete equipment shall be furnished (see 6.3.2).

3.1.2 Compliance with specified requirements.-

3.1.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.1.2.3 for exception).

3.1.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.1.4) preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.1.2.3 When deviation from 3.1.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:05Z
Check the source to verify that this is the current version before use.

3.1.3 Test data.- The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like.)

3.1.4 Reference standards.- After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.1.5 Preproduction inspection.- Preproduction samples may be submitted by the Government to the group A inspection specified in table II and to the preproduction inspection specified in table I. Unlisted nondestructive inspection on preproduction samples of equipment and unlisted destructive or nondestructive inspection on separate preproduction samples of parts or materials may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order: (1) bounce, (2) shock: drop; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table I - Preproduction inspection

Inspection	Requirement Paragraph	Test Paragraph
Bounce test; light equipment	3.6	4.4
Shock: drop test	3.6.1	4.5
Temperature test	3.5	4.6
Moisture-resistance test	3.5	4.7

3.2 Riveting.- After riveting, the joints shall be tight, the jointed parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.3 Cleaning.- Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no visible or latent deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.4 Construction.- The theodolite mount shall be constructed in accordance with Drawing and Data List SC-DL-86479.

3.5 Service conditions.- The theodolite mount shall meet the following service conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:05Z
Check the source to verify that this is the current version before use.

3.5.1 Equipment operating The theodolite mount shall meet the requirements of this specification under any probable combination of the following:

(a) Temperature. Any ambient temperature in the range of -40°F. to +150°F., as determined by the test of 4.6.1.

(b) Relative humidity. Any relative humidity up to 95 percent, as determined by the test of 4.7.

3.5.2 Equipment non-operating The mount shall meet the requirements of 3.5.1 after subjection to any one or more of the following non-operating conditions:

(a) Temperature. Continuous exposure for 24 hours at +160°F., and 24 hours at -80°F., as determined by the test of 4.6.2.

(b) Relative humidity. Relative humidity up to 100 percent, including condensation caused by temperature changes, as determined by the test of 4.7.

(c) Bounce and shock: drop. See 3.6.

3.5.3 Following subjection of the temperature and humidity tests of 3.5.1 and 3.5.2 respectively, the paint finish on the mount shall not crack, blister, peel, or show any other defect.

3.6 Bounce The equipment shall show no physical damage except surface abrasion, and shall be completely operable after being subjected to the test of 4.4.

3.6.1 Shock: drop The equipment shall show no physical damage except surface abrasion, and shall be completely operable after being subjected to the test of 4.5.

3.7 Running spare parts Running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2.)

3.8 Bounce conditioning After final assembly, each component on contract, with shock mounts removed or blocked, shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this conditioning for 1 minute. After conditioning, the equipment shall be capable of meeting the inspection of section 4.

3.9 Operational After assembly, the theodolite mount shall be operated in order to determine whether it is adjustable to any height within the specified limits covered by Drawing and Data List SC-DL-86479. The lamp for the recording shelf shall be capable of providing light by connecting its line cord to a suitable power supply and operating the lamp switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:05Z
Check the source to verify that this is the current version before use.

3.10 Workmanship Components, subassemblies and parts shall be manufactured and assembled in a thoroughly workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection, general Government inspection shall consist of group A inspection as specified in 4.1.1 (see 6.3.1.)

4.1.1 Group A inspection Group A inspection shall be in accordance with table II and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector, except that the operational examination (see 4.3) shall be last.

Table II - Government group A inspection

Inspection	Requirement Paragraph	Test Paragraph	AQL
Visual and mechanical, major		4.2	1 %
Visual and mechanical, minor		4.2	4 %
Operational	3.9	4.3	1%

NOTE: Prior to acceptance inspection, the equipment shall meet the requirements of 3.8.

4.1.2 Disposition of nonconforming product When defective sample units or rejected lots are resubmitted for acceptance, such items shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the items are resubmitted and shall become the property of the Government.

4.2 Visual and mechanical inspection. Parts and equipment shall be inspected for workmanship, mechanical fit, loose nuts and screws, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Wiring, soldered connections, ground connections, welds, finishes, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of Standard MIL-STD-105.

4.3 Operational test The theodolite mount shall be operated to determine compliance with the requirements of 3.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:05Z
Check the source to verify that this is the current version before use.

4.4 Bounce test; light equipment The theodolite mount, as packaged for shipment, shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 r.p.m. ± 1 percent, for a total of 3 hours. During each 1/2-hour period of the test, the packaged equipment shall rest on a different face.

4.5 Shock: drop test The equipment, as packaged for shipment, shall be dropped a distance of 30 inches on each face, edge and corner (total of 26 drops) in the order indicated. Should failure of the container occur after any drop subsequent to the 8th drop, the test shall be discontinued. The equipment shall then be removed from the container and upon examination, may show minor mechanical damage but shall be completely operable. Drops shall be made from a quick-release hook; and Acme Drop Tester, or equal; a Conbur Tester (10-degree inclined plane), or equal, or a basket-type pendulum tester. The floor or barrier receiving the impact shall be solid steel or concrete.

4.6 Temperature test The theodolite mount shall be subjected to the following operating and non-operating temperature tests.

4.6.1 Operating The equipment shall be kept at -40°F. for a period of 8 hours. At the end of this period, and while at this temperature, the equipment shall be operable. The equipment shall then be subjected to a temperature of +150°F. for a period of 8 hours. At the end of this period, and while at this temperature, the equipment shall be operable.

4.6.2 Non-operating The equipment shall be kept at -80°F. for 24 hours and then returned to +77 ±9°F. for 4 hours. The equipment shall then be held at +160°F. for 24 hours and then returned to +77 ±9°F. for 4 hours. At the end of this cycle, the equipment shall be operable, and shall meet the requirements of 3.5.3 for paint finish.

4.7 Moisture-resistance The equipment shall be subjected to 5 continuous cycles of the test shown on Drawing SC-D-15914. After completion of the final cycle, the equipment shall be operable, and shall meet the requirements of 3.5.3 for paint finish.

5 PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 For domestic shipment Theodolite Mount ML-180-( ) shall be preserved and packaged in accordance with Method III of Specification MIL-P-116 (see 6.2.)

5.1.2 For export shipment Preserve and package Theodolite Mount ML-180-( ) in accordance with procedures specified for the designated methods as prescribed in Specification MIL-P-116, and further described in the following paragraphs: (see 6.2)

5.1.2.1 Electrical spare parts items Package each electrical spare parts item, determined in quantity and number from the bid request or contract, in accordance with the applicable methods described in Specification MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:05Z
Check the source to verify that this is the current version before use.

5.1.2.2 Consolidated package Consolidate the items, packaged as specified in 5.1.2.1, together within a suitable style and grade corrugated fiberboard box. Fill all voids with flexible corrugated paper to prevent movement. Seal box closure with suitable tape, adhesive or stapling.

5.1.2.3 Theodolite Mount ML-180-( ) Package each theodolite mount Method IC-5, as follows: The assembled, complete theodolite mount shall be adequately blocked and braced by means of a blocking assembly, fabricated of wood panels and wood blocks to bridge projections or voids and to square the unit up for proper packaging. Any surfaces of the assembly contacting the components shall be covered with suitable cushioning material to prevent abrasion. Place and secure consolidated package as specified in 5.1.2.2 within blocking assembly with suitable tape. Cushion each item on all surfaces with cells or pads or both fabricated of corrugated fiberboard, conforming to Specification 19-10 and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned item within a close-fitting suitable style and grade fiberboard box conforming to Specification JAN-P-108. Seal entire box closure, corners and joints with water-resistant pressure-sensitive tape, conforming to Specification JAN-P-127.

5.2 Packing

5.2.1 For domestic shipment Theodolite Mount ML-180-( ) shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood, or corrugated fiberboard boxes conforming to Specification NN-B-591, NN-B-601, NN-B-621, and LLL-B-631, respectively. Closures shall be made in accordance with the applicable box specifition. (See 6.2.)

5.2.2 For export shipment Theodolite Mount ML-180-( ), packaged as specified in 5.1.2.3 shall be packed in a wood cleated plywood, nailed wood or wire bound wood box conforming to the requirements of Specifications JAN-P-105, JAN-P-106 and MIL-B-107, respectively. Fabricate the boxes to fit contents snugly. Close the shipping container as specified in the appendix of the applicable box specification. (See 6.2.)

5.2.2.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment of ports.

5.3 Marking for shipment Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of Standard MIL-STD-129.

6 NOTES

6.1 Intended use Theodolite Mount ML-180-( ) is intended for use in supporting Theodolite ML-247.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:05Z
Check the source to verify that this is the current version before use.

6.2 Ordering data Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether domestic or overseas shipment is required. (See 5.1.1,5.1.2, 5.2.1 and 5.2.2)

(c) Samples required. (See 3.1)

(d) Marking and shipping of samples.

(e) Quantity of running spare parts required. (See 3.7)

6.3 Contractual requirements It is recommended that the following be included in the contract:

6.3.1 Additional testing It should be understood that additional inspection and nondestructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable documents. The Government inspector, therefore, may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection and testing, performed either at a Government laboratory or the contractor's plant.

6.3.2 Waiver of preproduction samples. When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that the specified samples have been submitted and approved on the current contract; production of the item under the new order will be concurrent with, or following without interruption, production on the current order; and current production is in strict compliance with specified requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

6.4 Nomenclature The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example, ML-180-W. As soon as possible after the award of the contract, the contractor should apply to the contracting officer for such information. (See 1.1.)

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 20260-17T23:05Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-0879

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:05Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:05Z
Check the source to verify that this is the current version before use.